

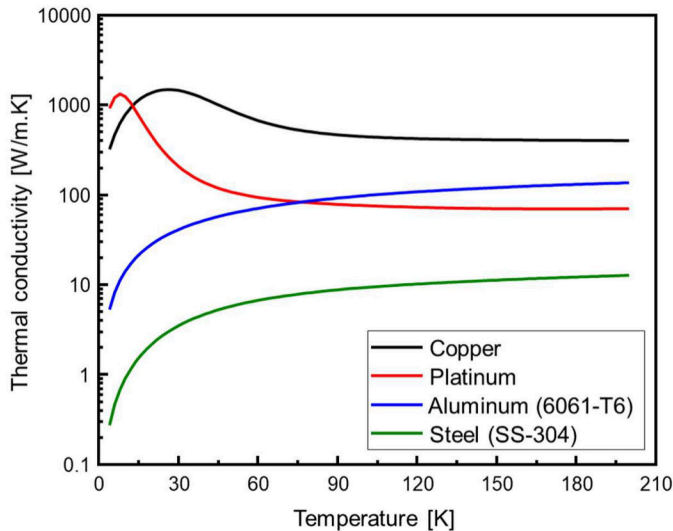


Fig. 1. Variation of thermal conductivity of metals with temperature (taken from NIST database [10]).

2.2. Natural convection (NC)

2.3. Nucleate boiling (NB)

Once the onset of nucleate boiling (ONB) is reached (i.e., ΔT_{ONB} , $q'' = q''_{ONB}$), bubble formation begins, and the heat transfer coefficient (HTC) increases significantly during the nucleate boiling (NB) regime. The HTC remains relatively constant until it begins to subside as the flow transitions to the fully developed nucleate boiling (FNB) regime. Based on the authors' previous studies, a baseline correlation for the HTC was developed [11], which was validated against experimental data for cryogenic saturated conditions. The correlation accounts for heater orientation, size, material, and the form of the baseline correlation.

$$h_{nb} = 13.3 q''^{0.665} (1 + 0.52 \dots)$$

where h_{nb} is the NB HTC [W/m²·K], q'' is the heat flux provided to the heater [W/m²], and Pr is the Prandtl number. Details concerning the correlation and its predictive performance are provided in [11].

In the present study, an experimental investigation was conducted to investigate the effect of material properties on the HTC. As an initial screening study, materials other than copper were excluded due to the lack of experimental data for the